



BIOLOGICAL SYSTEMS MODELING AND ANALYSIS

Term Project

【Spring Semester, 2025】

Due Date: June 15 (Monday), 2026 before 17:00

PART 1: (50%)

Complete exercise problems 1~7 of Chapter 12 “Hormonal Control in Mammals” of our textbook after reading the contents of Chapter 12 and necessary papers or materials pertaining to the modeling of glucose and insulin regulation.

PART 2: (50%)

Read Chapter 15 “Disease” for the background knowledge you need for the following tasks.

Examine the effects of a HIV vaccination on the sIC AIDS model (loosely based on Garnett et al. (2002)). Assume that the vaccine is applied to $S_{f,2}$ and $S_{m,2}$ at rate $v = 0.65/\text{year}$ per individual. Vaccinated individuals become protected (i.e, move into variables $P_{f,2}$ and $P_{m,2}$). Further assume that a fraction of vaccinated individuals lose protection at rate $l = 0.1/\text{year}$.

Address the following questions:

- Will vaccination cause the epidemic to peak, and if so, when will the decline occur?
- What will be the total cost of vaccination (assuming each vaccination costs \$10)?
- What is the optimum vaccination rate needed to effectively control the epidemic?
- Which intervention strategy is more effective: vaccination or safe-sex education and condom use?

Support your discussion and conclusions with simulation results and analyses based on the sIC AIDS model.



Report Preparation Guideline:

- Write a report describing in details of your work in solving the problems in Part 1 and Part 2.
- You will need to give a 15-minute talk in the class on **June 11** to present your work on Part 2 of this term project.
- The materials you need to submit for your term project are as follows (submit these materials to the NTU COOL course website):
 1. Project report (including figures and tables)
 2. Simulation programs
 3. Papers you have read in preparing your term project
 4. Presentation powerpoint file

References:

Part 1:

1. Cobelli, C. G. Federspil, G. Pacini, A. Salvan, and C. Scandellari. 1982. An integrated mathematical model of the dynamics of blood glucose and its hormonal control. *Mathematical Biosciences* 58: 27-60.
2. Makroglou, A. J. Li, and Y. Kuang. 2006. Mathematical models and software tools for the glucose-insulin regulatory system and diabetes: an overview. *Applied Numerical Mathematics* 56(3-4): 559-573.
3. Man, C. D., R. A. Rizza, and C. Cobelli. 2007. Meal simulation model of the glucose-insulin system. *IEEE Trans. on Biomed. Eng.* 54(10): 1740-1749.

Part 2:

4. Garnett, G. P. and R. M. Anderson. 1993. Factors controlling the spread of HIV in heterosexual communities in developing countries: patterns of mixing between different age and sexual activity classes. *Philosophical Transactions of the Royal Society, London*, B 342:137-159.
5. Garnett, G. P., K. Desai, and J. Williams. 2002. Technical Annex I. The epidemiological impact of an HIV/AIDS vaccination as a function of vaccine properties: Results of the Imperial College model. Pages i-26 in J. Stover, G. P.



Garnett, S. Seitz, and S. Forsythe, editors. Policy Research Working Paper 2811.
World Bank, World Bank, New York, New York, USA. URL:

<https://documents.worldbank.org/en/publication/documents-reports/documentdetail/896881468760559617/the-epidemiological-impact-of-an-hiv-aids-vaccine-in-developing-countries>